

Networks and Systems

BSc Computer Science - Durham University

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1 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

1.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Cyber Security	10/10/2026	06/11/2026	04/12/2026	Dr Carlton Shepherd	25%
Networks	31/10/2026	08/01/2027	05/02/2027	Dr Wanqing Tu	25%
Dist. Sys. & Databases	May Exam	n/a	n/a	Dr Vasantam	50%

Table 1: Assessments for Networks and Systems.

1.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Thursday	09:00 - 11:00	Lecture	CLC202	Dr Carlton Shepherd
Thursday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 2: Michaelmas Term Timetable for Networks and Systems.

2 Cyber Security

This section is taught by Dr Carlton Shepherd. **Check for recommended reading.**

2.1 Introduction, Threat Modelling, and Applied Cryptography

3 Networks

This section is taught by Dr Wanqing Tu. **Check for recommended reading.**

4 Databases

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

- 4.1 Enhanced Entity-Relationship Model**
- 4.2 Semistructured Databases (XML)**
- 4.3 XML Data Manipulation (XPath and XQuery)**
- 4.4 Transactions**
- 4.5 Concurrency Control**
- 4.6 Distributed Transactions**
- 4.7 Distributed Concurrency Control**

5 Distributed Systems

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].

- 5.1 Introduction to Distributed Systems**
- 5.2 Middleware and RMI Technologies**
- 5.3 Message-Oriented Middleware and Web Services**
- 5.4 Replication Models**
- 5.5 Concurrent Events and Consistency Control**
- 5.6 Fault Tolerance**
- 5.7 Load Distribution**

Part I

Appendix

6 Bibliography

6.1 Module Professors

Module Lead: Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: Thursday 11:00 - 12:00.

Lecturer: Doctor Thirupathaiah Vasantam

- Focus: Applied probability and performance modelling.
- Email: thirupathaiah.vasantam@durham.ac.uk
- Office Hours: Monday 14:00 - 15:00.

Lecturer: Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: Email to make an appointment.

6.2 References

6.3 List of Figures

6.4 Reading List

6.4.1 Databases

Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

6.4.2 Distributed Systems

Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].